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Hypoxia and mTOR Inhibition Reshape the Transcriptome of Normal and Triple Negative Breast Cancer Cells

A multifactor RNA seq differential expression analysis (GSE104193)

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1. Background and Motivation

Hypoxia is a defining feature of the tumour microenvironment. Cancer cells frequently alter their transcriptional and translational programmes to survive under low oxygen conditions. The mTOR pathway is a central regulator of protein synthesis and is closely linked to oxygen and nutrient sensing. This makes mTOR inhibitors, such as PP242, of interest both as research tools and as potential cancer therapeutics.

This project analyses a public RNA seq dataset (GSE104193 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE104193>)), comparing a non malignant breast epithelial cell line (MCF10A) with a triple negative breast cancer cell line (MDA-MB-231). Cells were exposed to normoxia or hypoxia, with and without the mTORC1/2 inhibitor PP242, and both total and polysome bound mRNA were profiled. This gives a factorial design (2 cell lines x 2 oxygen levels x 2 drug conditions x 2 mRNA fractions, 2 biological replicates each, 32 samples total), which was used to ask:

1. How does hypoxia change gene expression, and does this differ between the normal and cancer cell lines?
2. Is hypoxia's effect on transcription also seen proportionally in the polysome bound (translating) mRNA pool, or is there evidence of hypoxia specific translational control?

2. Data and Methods

2.1 Data retrieval

I retrieved the raw RSEM expected counts and sample metadata directly from GEO using the `GEOquery` package. Two supplementary count files were provided, split by cell line, which I merged into a single 60,554 gene by 32 sample count matrix. Sample identity across the two files could not be assumed from row or column order, so this was resolved using the explicit `barcode:ch1` metadata field instead. I checked the mapping was correct by confirming exact agreement between the metadata sample IDs and the count matrix column names before proceeding.

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```
gse <- getGEO("GSE104193", GSEMatrix = TRUE)
gse <- gse[[1]]
pheno <- pData(gse)

supp_dir <- "data/raw/GSE104193"
supp_files_expected <- file.path(supp_dir, c(
  "GSE104193_COUNTS_POLYSOME_01_new_pipeline.txt.gz",
  "GSE104193_COUNTS_POLYSOME_02_new_pipeline.txt.gz"
))

if (!all(file.exists(supp_files_expected))) {
  getGEOSuppFiles("GSE104193", baseDir = "data/raw")
} else {
  message("Supplementary files already present locally – skipping re-d
ownload.")
}

counts_01 <- read.delim(supp_files_expected[1],
  header = TRUE, row.names = 1, check.names = F
  ALSE)
counts_02 <- read.delim(supp_files_expected[2],
  header = TRUE, row.names = 1, check.names = F
  ALSE)
```

2.2 Sample metadata construction

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```
sample_info <- pheno %>%
  rownames_to_column("geo_accession_row") %>%
  transmute(
    gsm = geo_accession,
    barcode = `barcode:ch1`,
    cell_line = `cell line:ch1`,
    fraction = `fraction:ch1`,
    replicate = `replicate:ch1`,
    treatment_raw = `treatment:ch1`
  ) %>%
  separate(treatment_raw, into = c("oxygen", "drug"), sep = "\\+", fill = "right") %>%
  mutate(
    drug = ifelse(is.na(drug), "Control", drug),
    fraction = recode(fraction, "Total mRNA" = "Total", "Polysomal mRNA" = "Polysomal"),
    replicate = recode(replicate, "Biological replicate 1" = "Rep1", "Biological replicate 2" = "Rep2")
  )

counts_combined <- cbind(counts_01, counts_02)
barcode_to_gsm <- setNames(sample_info$gsm, sample_info$barcode)
colnames(counts_combined) <- barcode_to_gsm[colnames(counts_combined)]
counts_combined <- counts_combined[, sample_info$gsm]

stopifnot(all(colnames(counts_combined) == sample_info$gsm))

coldata <- sample_info
rownames(coldata) <- coldata$gsm
coldata$cell_line <- factor(coldata$cell_line, levels = c("MCF10A", "MDA-MB-231"))
coldata$fraction <- factor(coldata$fraction, levels = c("Total", "Polysomal"))
coldata$oxygen <- factor(coldata$oxygen, levels = c("Normoxia", "Hypoxia"))
coldata$drug <- factor(coldata$drug, levels = c("Control", "PP242"))
coldata$replicate <- factor(coldata$replicate)

kable(head(coldata))
```

	gsm	barcode	cell_line	fraction	replicate	oxygen	drug
GSM2791560	GSM2791560	O097	MCF10A	Total	Rep1	Normoxia	Control
GSM2791561	GSM2791561	O098	MCF10A	Total	Rep1	Hypoxia	Control
GSM2791562	GSM2791562	O099	MCF10A	Total	Rep1	Normoxia	PP242
GSM2791563	GSM2791563	O100	MCF10A	Total	Rep1	Hypoxia	PP242
GSM2791564	GSM2791564	O101	MCF10A	Polysomal	Rep1	Normoxia	Control
GSM2791565	GSM2791565	O102	MCF10A	Polysomal	Rep1	Hypoxia	Control

2.3 Differential expression model

Counts were rounded to integers, as required by DESeq2, because RSEM produces expected counts rather than integer counts. Genes with fewer than 10 reads in at least 4 samples were removed before modelling.

I used a multi-factor design:

```
~ cell_line + oxygen + drug + fraction + cell_line:oxygen + oxygen:fraction
```

I did not fit a fully saturated four way factorial model. With only 2 biological replicates per condition, an unconstrained model would leave very few residual degrees of freedom for reliable dispersion estimation. Instead, the interaction terms included were chosen based

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on the specific questions in Section 1. The `cell_line:oxygen` term tests whether the hypoxia response differs between the normal and cancer cell lines. The `oxygen:fraction` term tests whether hypoxia's effect on transcription is also seen proportionally in the polysome bound (translating) mRNA pool.

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```
counts_int <- round(as.matrix(counts_combined))

dds <- DESeqDataSetFromMatrix(
  countData = counts_int,
  colData = coldata,
  design = ~ cell_line + oxygen + drug + fraction + cell_line:oxygen +
oxygen:fraction
)
keep <- rowSums(counts(dds) >= 10) >= 4
dds <- dds[keep, ]
dds <- DESeq(dds)

nrow(dds)
```

```
## [1] 17910
```

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```
resultsNames(dds)
```

```
## [1] "Intercept"                                "cell_line_MDA.MB.231_vs_MC
F10A"
## [3] "oxygen_Hypoxia_vs_Normoxia"              "drug_PP242_vs_Control"
## [5] "fraction_Polysomal_vs_Total"             "cell_lineMDA.MB.231.oxygen
Hypoxia"
## [7] "oxygenHypoxia.fractionPolysomal"
```

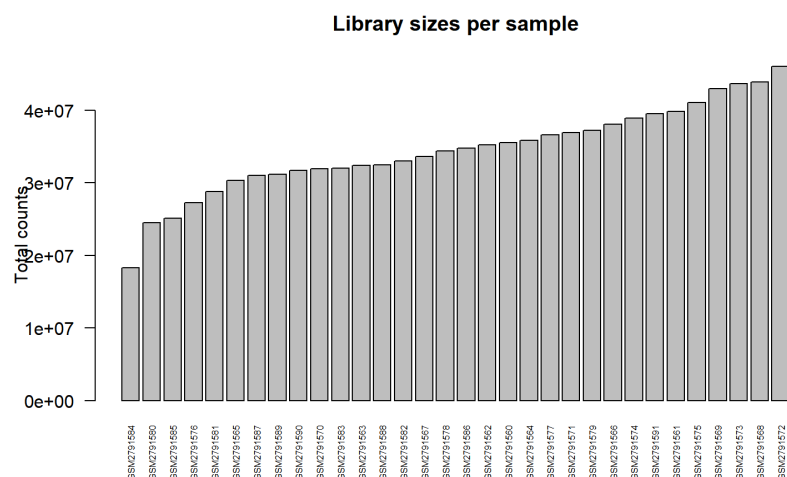
Filtering retained 17910 of the original 60554 genes for modelling. Effect sizes were shrunk using `apeglm` (Zhu, Ibrahim and Love, 2018) before ranking and visualisation.

3. Quality Control

3.1 Library sizes

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```
library_sizes <- colSums(counts(dds))
barplot(sort(library_sizes), las = 2, cex.names = 0.5,
  main = "Library sizes per sample", ylab = "Total counts")
```



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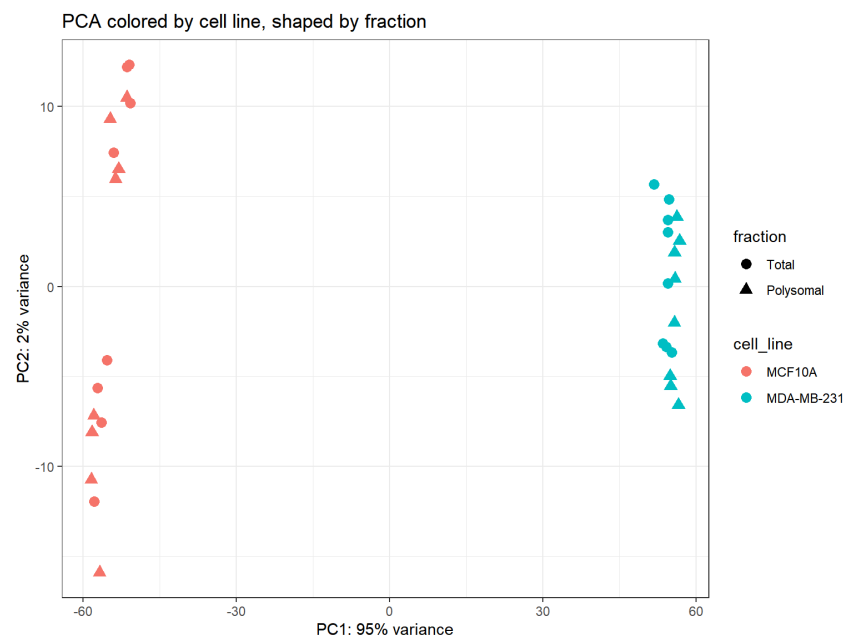
Library sizes ranged from 18.3M to 46M reads across the 32 samples. This was accounted for using DESeq2's median of ratios size factor normalisation.

3.2 Sample structure (PCA)

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```
vsd <- vst(dds, blind = TRUE)
pca_data <- plotPCA(vsd, intgroup = c("cell_line", "oxygen", "drug",
"fraction"), returnData = TRUE)
percentVar <- round(100 * attr(pca_data, "percentVar"))

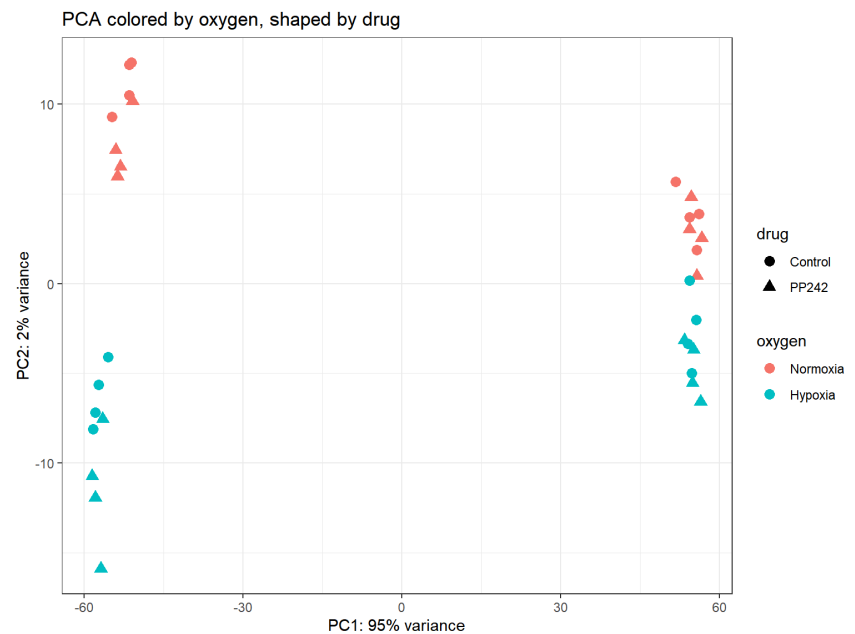
ggplot(pca_data, aes(x = PC1, y = PC2, color = cell_line, shape = frac
tion)) +
  geom_point(size = 3) +
  xlab(paste0("PC1: ", percentVar[1], "% variance")) +
  ylab(paste0("PC2: ", percentVar[2], "% variance")) +
  theme_bw() +
  labs(title = "PCA colored by cell line, shaped by fraction")
```



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```
ggplot(pca_data, aes(x = PC1, y = PC2, color = oxygen, shape = drug))
+
  geom_point(size = 3) +
  xlab(paste0("PC1: ", percentVar[1], "% variance")) +
  ylab(paste0("PC2: ", percentVar[2], "% variance")) +
  theme_bw() +
  labs(title = "PCA colored by oxygen, shaped by drug")
```

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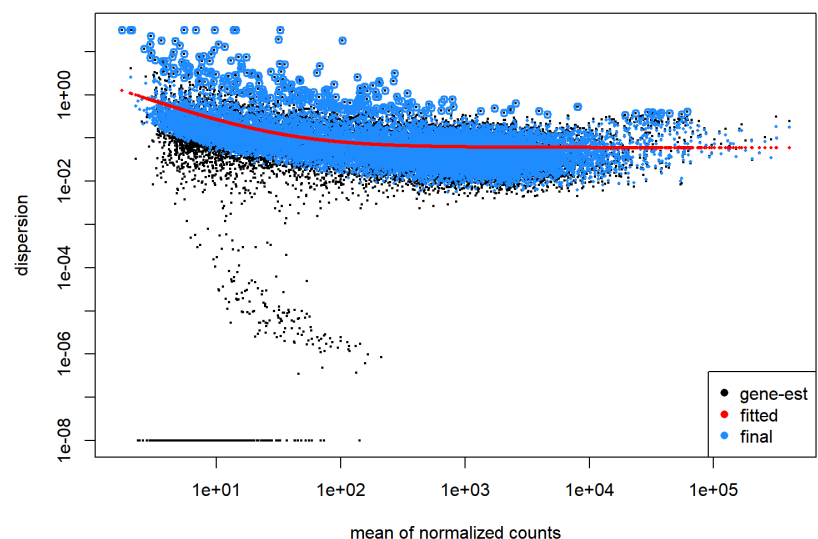


PC1 (95% of variance) separated samples by cell line. This is expected, as MCF10A (non malignant) and MDA-MB-231 (triple negative cancer) are very different cell types at the transcriptional level. PC2 (2% of variance) separated samples by oxygen level within both cell lines. Although this is a small proportion of the total variance, the separation was consistent across both cell lines, which suggests a real hypoxia signature. Neither drug nor mRNA fraction produced a visible separation on the top two PCs. This suggests these effects are smaller, and may be restricted to specific groups of genes rather than affecting the whole transcriptome.

3.3 Dispersion estimates

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```
plotDispEsts(dds)
```



Gene wise dispersion estimates shrank towards the fitted trend, as expected. A small number of genes showed unusually low dispersion estimates at low mean counts. This is a known artefact caused by rounding the non integer RSEM expected counts to integers, and is corrected by DESeq2's shrinkage.

4. Results

4.1 Hypoxia response in normal (MCF10A) cells

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```
res_hypoxia_MCF10A <- results(dds, name = "oxygen_Hypoxia_vs_Normoxi
a")
res_hypoxia_MCF10A_shrunk <- lfcShrink(dds, coef = "oxygen_Hypoxia_vs_
Normoxia", type = "apeglm")
summary(res_hypoxia_MCF10A)
```

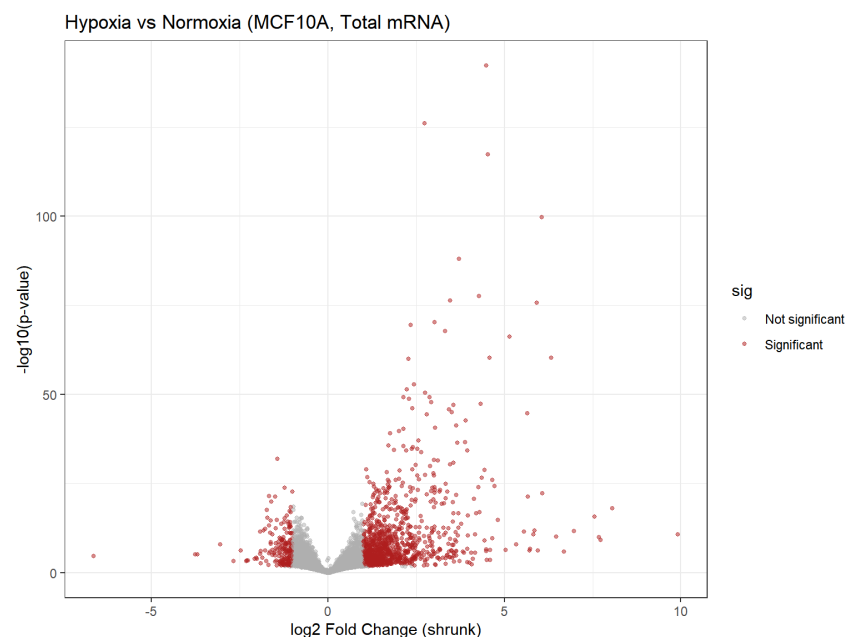
```
##
## out of 17910 with nonzero total read count
## adjusted p-value < 0.1
## LFC > 0 (up)      : 2965, 17%
## LFC < 0 (down)    : 2406, 13%
## outliers [1]      : 0, 0%
## low counts [2]    : 0, 0%
## (mean count < 2)
## [1] see 'cooksCutoff' argument of ?results
## [2] see 'independentFiltering' argument of ?results
```

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```
gene_info <- data.frame(
  full_id = rownames(dds),
  ensembl_id = sapply(strsplit(rownames(dds), ","), `[`, 1),
  gene_symbol = sapply(strsplit(rownames(dds), ","), `[`, 2),
  biotype = sapply(strsplit(rownames(dds), ","), `[`, 3)
)

volcano_df <- as.data.frame(res_hypoxia_MCF10A_shrunk)
volcano_df$gene_symbol <- gene_info$gene_symbol[match(rownames(volcano
_df), gene_info$full_id)]
volcano_df$sig <- ifelse(!is.na(volcano_df$padj) & volcano_df$padj <
0.05 & abs(volcano_df$log2FoldChange) > 1,
  "Significant", "Not significant")

ggplot(volcano_df, aes(x = log2FoldChange, y = -log10(pvalue), color =
sig)) +
  geom_point(alpha = 0.5, size = 1) +
  scale_color_manual(values = c("Significant" = "firebrick", "Not sign
ificant" = "grey70")) +
  theme_bw() +
  labs(title = "Hypoxia vs Normoxia (MCF10A, Total mRNA)",
    x = "log2 Fold Change (shrunk)", y = "-log10(p-value)")
```

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```
top_hypoxia_genes <- volcano_df %>%
  filter(!is.na(padj)) %>%
  arrange(padj) %>%
  dplyr::select(gene_symbol, log2FoldChange, padj) %>%
  head(20)

kable(top_hypoxia_genes, digits = 3)
```

	gene_symbol	log2FoldChange	padj
ENSG00000181458.10,TMEM45A,protein_coding	TMEM45A	4.483	0
ENSG00000111674.8,ENO2,protein_coding	ENO2	2.731	0
ENSG00000147852.15,VLDLR,protein_coding	VLDLR	4.535	0
ENSG00000107159.12,CA9,protein_coding	CA9	6.060	0
ENSG00000113083.12,LOX,protein_coding	LOX	3.712	0
ENSG00000061656.9,SPAG4,protein_coding	SPAG4	4.276	0
ENSG00000074410.13,CA12,protein_coding	CA12	3.461	0
ENSG00000129521.13,EGLN3,protein_coding	EGLN3	5.910	0
ENSG00000170379.19,TCAF2,protein_coding	TCAF2	3.020	0
ENSG00000104765.14,BNIP3L,protein_coding	BNIP3L	2.352	0
ENSG00000114268.11,PFKFB4,protein_coding	PFKFB4	3.314	0
ENSG00000143847.15,PPFIA4,protein_coding	PPFIA4	5.145	0
ENSG00000165507.8,C10orf10,protein_coding	C10orf10	6.333	0
ENSG00000205129.8,C4orf47,protein_coding	C4orf47	4.580	0
ENSG00000072682.18,P4HA2,protein_coding	P4HA2	2.286	0
ENSG00000163710.7,PCOLCE2,protein_coding	PCOLCE2	2.434	0
ENSG00000102144.13,PGK1,protein_coding	PGK1	2.239	0
ENSG00000133874.1,RNF122,protein_coding	RNF122	2.758	0
ENSG00000159167.11,STC1,protein_coding	STC1	2.880	0
ENSG00000122884.12,P4HA1,protein_coding	P4HA1	2.146	0

Hypoxia caused widespread transcriptional changes in MCF10A cells. The genes with the strongest induction included known HIF target genes: CA9 and CA12 (pH regulation), EGLN3 (part of the oxygen sensing feedback loop), BNIP3L (hypoxia induced mitophagy), and the glycolysis genes PFKFB4 and PGK1. This shows the pipeline correctly identifies known hypoxia biology, which supports the validity of the wider analysis.

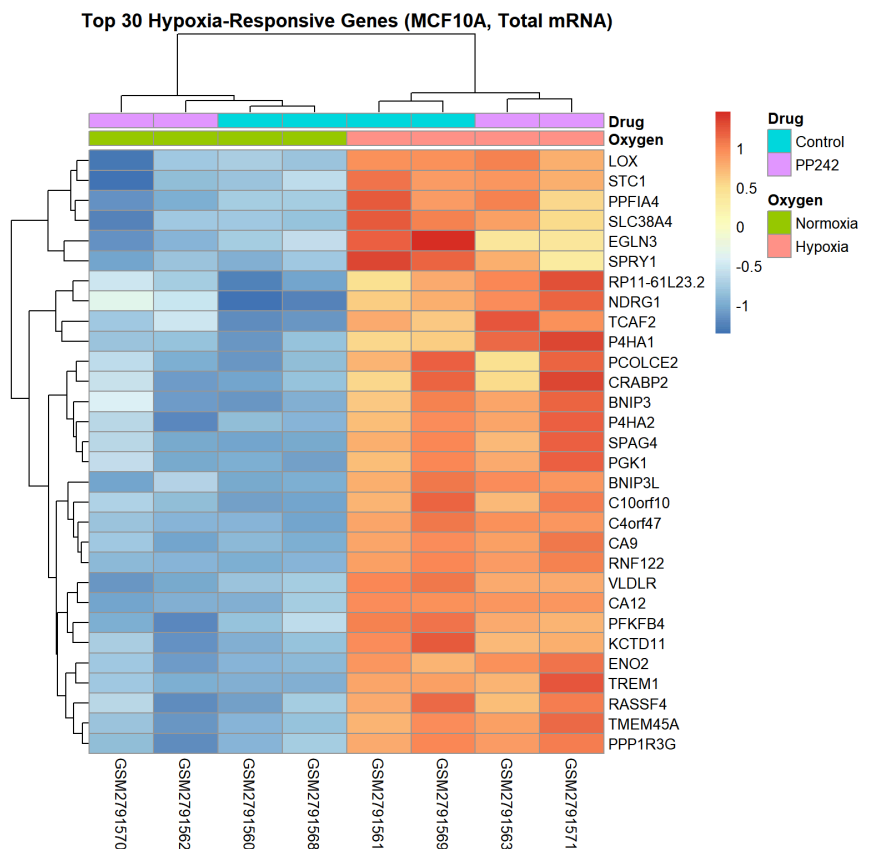
Hide

```
top30 <- volcano_df %>% filter(!is.na(padj)) %>% arrange(padj) %>% head(30)
mcf10a_total_samples <- rownames(coldata)[coldata$cell_line == "MCF10A" & coldata$fraction == "Total"]
mat <- assay(vsd)[rownames(top30), mcf10a_total_samples]
mat_scaled <- t(scale(t(mat)))
rownames(mat_scaled) <- top30$gene_symbol

annotation_col <- data.frame(
  Oxygen = coldata[mcf10a_total_samples, "oxygen"],
  Drug = coldata[mcf10a_total_samples, "drug"]
)
rownames(annotation_col) <- mcf10a_total_samples

pheatmap(mat_scaled, annotation_col = annotation_col,
  main = "Top 30 Hypoxia-Responsive Genes (MCF10A, Total mRNA)")
```

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4.2 Hypoxia response differs between normal and cancer cells

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```
res_interaction_cellline_oxygen <- results(dds, name = "cell_lineMDA.M
B.231.oxygenHypoxia")
res_interaction_cellline_oxygen_shrunk <- lfcShrink(dds, coef = "cell_
lineMDA.MB.231.oxygenHypoxia", type = "apeglm")
res_hypoxia_MDAMB231 <- results(dds, contrast = list(c("oxygen_Hypoxia
_vs_Normoxia", "cell_lineMDA.MB.231.oxygenHypoxia")))

summary(res_interaction_cellline_oxygen)
```

```
##
## out of 17910 with nonzero total read count
## adjusted p-value < 0.1
## LFC > 0 (up)      : 1264, 7.1%
## LFC < 0 (down)    : 1669, 9.3%
## outliers [1]      : 0, 0%
## low counts [2]    : 0, 0%
## (mean count < 2)
## [1] see 'cooksCutoff' argument of ?results
## [2] see 'independentFiltering' argument of ?results
```

Hide

```
summary(res_hypoxia_MDAMB231)
```

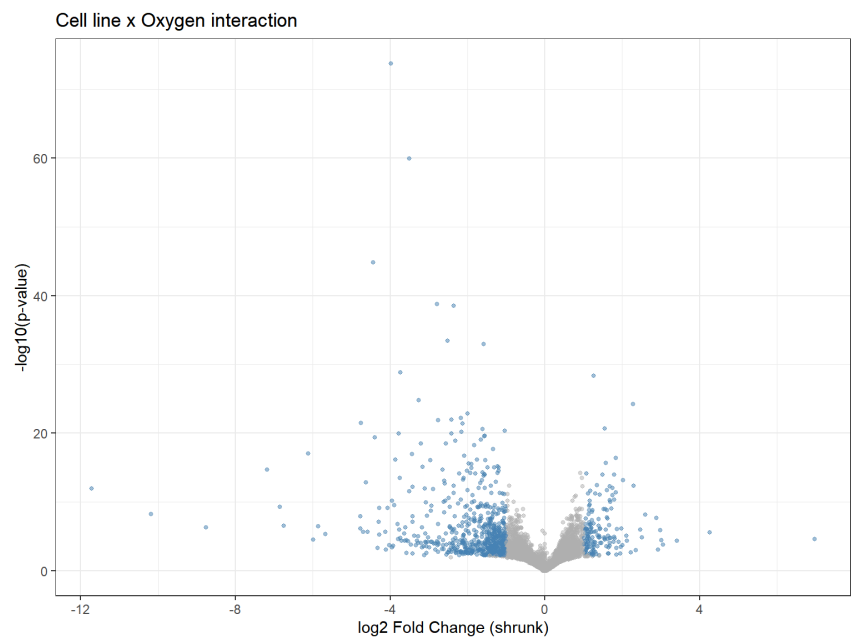
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```
##
## out of 17910 with nonzero total read count
## adjusted p-value < 0.1
## LFC > 0 (up)      : 814, 4.5%
## LFC < 0 (down)    : 528, 2.9%
## outliers [1]      : 0, 0%
## low counts [2]    : 0, 0%
## (mean count < 2)
## [1] see 'cooksCutoff' argument of ?results
## [2] see 'independentFiltering' argument of ?results
```

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```
interaction_df <- as.data.frame(res_interaction_cellline_oxygen_shrun
k)
interaction_df$gene_symbol <- gene_info$gene_symbol[match(rownames(int
eraction_df), gene_info$full_id)]

ggplot(interaction_df, aes(x = log2FoldChange, y = -log10(pvalue),
                           color = !is.na(padj) & padj < 0.05 & abs(l
og2FoldChange) > 1)) +
  geom_point(alpha = 0.5, size = 1) +
  scale_color_manual(values = c("TRUE" = "steelblue", "FALSE" = "grey7
0"), guide = "none") +
  theme_bw() +
  labs(title = "Cell line x Oxygen interaction",
       x = "log2 Fold Change (shrunk)", y = "-log10(p-value)")
```

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```
top_interaction_genes <- interaction_df %>%
  filter(!is.na(padj)) %>%
  arrange(padj) %>%
  dplyr::select(gene_symbol, log2FoldChange, padj) %>%
  head(20)

mdamb231_df <- as.data.frame(res_hypoxia_MDAMB231)
mdamb231_df$gene_symbol <- gene_info$gene_symbol[match(rownames(mdamb231_df), gene_info$full_id)]

comparison_table <- top_interaction_genes %>%
  dplyr::select(gene_symbol, interaction_LFC = log2FoldChange, interaction_padj = padj) %>%
  left_join(volcano_df %>% dplyr::select(gene_symbol, MCF10A_hypoxia_LFC = log2FoldChange), by = "gene_symbol") %>%
  left_join(mdamb231_df %>% dplyr::select(gene_symbol, MDAMB231_hypoxia_LFC = log2FoldChange), by = "gene_symbol")

kable(comparison_table, digits = 3)
```

gene_symbol	interaction_LFC	interaction_padj	MCF10A_hypoxia_LFC	MDAMB231_hypoxia_LFC
CA12	-3.984	0	3.461	
VLDLR	-3.510	0	4.535	
SPRY1	-4.438	0	4.325	
SLC37A2	-2.793	0	2.382	
PCOLCE2	-2.361	0	2.434	
PLXNB3	-2.517	0	2.225	
ENO2	-1.574	0	2.731	
EGLN3	-3.733	0	5.910	
RAB3IP	1.264	0	-1.011	
VWA1	-3.268	0	3.665	
TINAGL1	2.286	0	-1.727	
TMEM45A	-1.991	0	4.483	
STC1	-2.168	0	2.880	
TIMP3	-2.412	0	2.636	
PHEX	-2.755	0	1.921	
ANKFN1	-4.752	0	4.659	
SLC38A4	-2.129	0	2.800	
HSPH1	1.543	0	-0.707	
POLM	-1.615	0	1.351	
PARP8	-1.029	0	1.088	

The hypoxia response was significantly reduced in MDA-MB-231 compared with MCF10A. This is consistent with reports that triple negative breast cancer lines can already show a partially hypoxic phenotype under normoxic conditions, meaning there is less room for further induction.

The reduction was not the same for every gene. CA12, SPRY1 and ANKFN1 showed an almost complete loss of hypoxia induction in MDA-MB-231, while EGLN3 (PHD3) remained significantly induced in both cell lines, although more weakly in MDA-MB-231. This suggests the reduced response may be specific to downstream target genes, while the oxygen sensing machinery itself remains at least partly functional.

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4.3 Hypoxia's transcriptional effect and the translating mRNA pool

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```
res_interaction_oxygen_fraction <- results(dds, name = "oxygenHypoxia.fractionPolysomal")
summary(res_interaction_oxygen_fraction)
```

```
##
## out of 17910 with nonzero total read count
## adjusted p-value < 0.1
## LFC > 0 (up)      : 0, 0%
## LFC < 0 (down)    : 0, 0%
## outliers [1]      : 0, 0%
## low counts [2]    : 0, 0%
## (mean count < 2)
## [1] see 'cooksCutoff' argument of ?results
## [2] see 'independentFiltering' argument of ?results
```

No genes showed a significant oxygen:fraction interaction across the genome, even though hypoxia affected both the Total and Polysomal fractions individually and to a similar extent. This suggests the transcriptional response to hypoxia is passed through to the polysome bound mRNA pool at a similar level, with no strong evidence of hypoxia specific translational control in this dataset.

Limitation: with only 2 biological replicates per condition, this analysis does not have enough power to detect smaller or gene specific interaction effects. This result should be read as ruling out a large, genome wide effect, rather than ruling out translational regulation of individual transcripts.

4.4 Pathway-level validation (GSEA)

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```
gsea_input <- volcano_df %>%
  filter(!is.na(gene_symbol), !is.na(log2FoldChange)) %>%
  distinct(gene_symbol, .keep_all = TRUE)

gene_map <- bitr(gsea_input$gene_symbol, fromType = "SYMBOL", toType =
"ENTREZID", OrgDb = org.Hs.eg.db)

gsea_input <- gsea_input %>%
  inner_join(gene_map, by = c("gene_symbol" = "SYMBOL")) %>%
  distinct(ENTREZID, .keep_all = TRUE)

ranked_genes <- gsea_input$log2FoldChange
names(ranked_genes) <- gsea_input$ENTREZID
ranked_genes <- sort(ranked_genes, decreasing = TRUE)

hallmark_sets <- msigdb(species = "Homo sapiens", category = "H") %>%
  dplyr::select(gs_name, entrez_gene)

set.seed(42)
gsea_result <- GSEA(geneList = ranked_genes, TERM2GENE = hallmark_sets,
pvalueCutoff = 0.05, eps = 0)

gsea_df <- as.data.frame(gsea_result) %>% arrange(desc(NES))
kable(gsea_df[, c("ID", "NES", "pvalue", "p.adjust")], digits = 3, row.names = FALSE)
```

ID	NES	pvalue	p.adjust
HALLMARK_HYPOXIA	2.456	0.000	0.000
HALLMARK_GLYCOLYSIS	2.055	0.000	0.000

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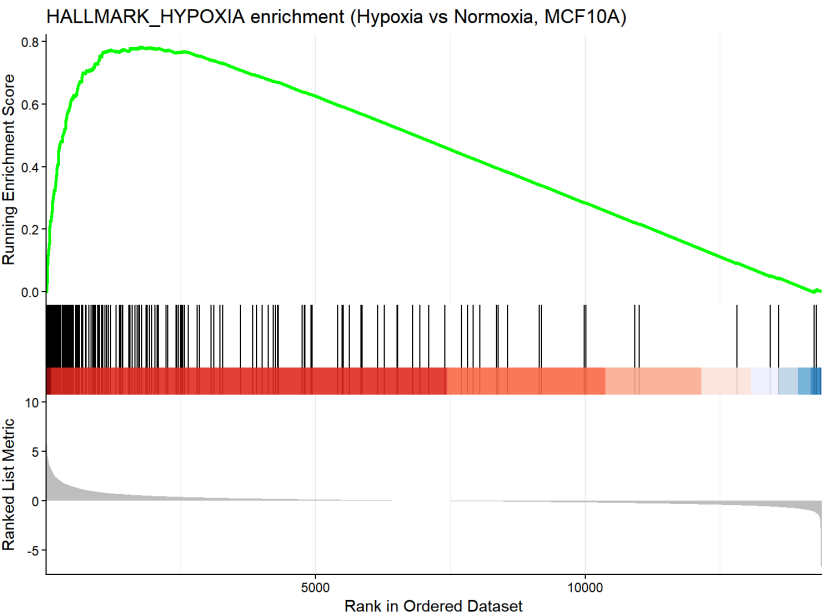
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ID	NES	pvalue	p.adjust
HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION	1.965	0.000	0.000
HALLMARK_INFLAMMATORY_RESPONSE	1.766	0.000	0.000
HALLMARK_TNFA_SIGNALING_VIA_NFKB	1.749	0.000	0.000
HALLMARK_KRAS_SIGNALING_UP	1.730	0.000	0.000
HALLMARK_COAGULATION	1.597	0.003	0.010
HALLMARK_MYOGENESIS	1.569	0.002	0.008
HALLMARK_APICAL_JUNCTION	1.544	0.001	0.004
HALLMARK_IL2_STAT5_SIGNALING	1.534	0.001	0.005
HALLMARK_UV_RESPONSE_DN	1.517	0.006	0.015
HALLMARK_ESTROGEN_RESPONSE_EARLY	1.462	0.005	0.013
HALLMARK_P53_PATHWAY	1.408	0.009	0.023
HALLMARK_APOPTOSIS	1.395	0.019	0.044
HALLMARK_SPERMATOGENESIS	-1.691	0.001	0.003
HALLMARK_DNA_REPAIR	-1.807	0.000	0.000
HALLMARK_OXIDATIVE_PHOSPHORYLATION	-1.925	0.000	0.000
HALLMARK_MYC_TARGETS_V2	-2.491	0.000	0.000
HALLMARK_G2M_CHECKPOINT	-2.704	0.000	0.000
HALLMARK_MYC_TARGETS_V1	-2.866	0.000	0.000
HALLMARK_E2F_TARGETS	-3.157	0.000	0.000

Hide

```
gseaplot2(gsea_result, geneSetID = "HALLMARK_HYPOXIA",
          title = "HALLMARK_HYPOXIA enrichment (Hypoxia vs Normoxia, M
CF10A)")
```



HALLMARK_HYPOXIA was the most significantly enriched pathway, which supports the differential expression results at the pathway level. The running enrichment score (green line) shows the Hallmark Hypoxia genes concentrated among the most strongly hypoxia upregulated transcripts in the ranked list, confirming the enrichment.

The wider results also formed a consistent pattern: HALLMARK_GLYCOLYSIS was upregulated alongside hypoxia. This reflects the shift towards glycolytic metabolism that occurs under low oxygen conditions. HALLMARK_OXIDATIVE_PHOSPHORYLATION was

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downregulated, which fits with this shift, as cells rely less on oxidative phosphorylation when oxygen is limited. Pathways linked to cell proliferation (HALLMARK_E2F_TARGETS, HALLMARK_G2M_CHECKPOINT, HALLMARK_MYC_TARGETS_V1/V2) were also strongly suppressed. This fits with the known hypoxia induced slowdown in cell cycle progression.

5. Discussion and Limitations

The aim of this analysis was to characterise the transcriptional response to hypoxia in a non malignant breast cell line, and to determine whether this response differs in a triple negative cancer cell line, or between total and polysome bound mRNA.

The results show a clear hypoxia signature in MCF10A cells. This is supported at multiple levels: individual genes matched known HIF targets, the pathway level GSEA result was strongly significant, and the PCA and heatmap both showed clean separation by oxygen condition. MDA-MB-231 cells showed a much weaker version of this response, and this reduction appeared to affect downstream effector genes more than the oxygen sensing machinery itself.

A key strength of this analysis was the multi factor design, which made it possible to test several biological questions from a single model rather than running separate pairwise comparisons. This is a more statistically defensible approach given the limited number of replicates available.

Several limitations should be considered:

- Replication: only 2 biological replicates were available per condition, which limits statistical power, particularly for interaction terms. The null oxygen:fraction result should not be taken as proof that no translational regulation occurs, only that a large effect was not detected.
- RSEM expected counts: rounding non integer expected counts to integers introduced minor artefacts in the dispersion estimates at low counts. These were corrected by DESeq2's shrinkage step.
- PP242 (drug) effect: the transcriptional effect of the mTOR inhibitor was comparatively small in this dataset and was not the focus of the main analysis. A more targeted analysis, for example restricted to known mTOR sensitive transcripts or the translational efficiency ratio, could investigate this further.

Overall, this analysis shows that a multi factor RNA seq design can be used to characterise a known biological response (hypoxia) while also testing more specific questions about how that response differs between cell types and mRNA fractions, within a single reproducible pipeline.

6. References

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